

**Computer Science Department
First year of Bachelor's degree**

Chapter 4 - AI for Document Retrieval (2)

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Introduction

■ **Literature review** is a **key step** in any **scientific project**.

With the rise of **AI**, it is **now** possible to automate and optimize **many tasks** related to the **collection, analysis & synthesis** of scientific information.

This chapter :

- presents **AI-based tools**,
- explains **how to write intelligent queries**,
- automatically summarize **articles**, and
- critically evaluate the **sources suggested** by these **tools**.

Presentation of AI tools

Several platforms use **AI** to facilitate **documentary research** :

Elicit : allows you to automatically extract key information from scientific publications, generate summaries, and suggest relevant articles

Scopus AI : Integrated into the Scopus database, it helps analyze publications, identify scientific trends, and recommend articles for further exploration of a topic

Web of Science Research Assistant : This intelligent assistant suggests relevant articles, creates reference lists, and allows you to quickly filter results based on **quality**, **impact**, and **relevance**

Designing Smart Requests

To get the most out of **AI tools**, it is essential to know how to write **effective queries** :

- Define **precise** & **relevant keywords** for the **research topic**
- Use **advanced filters** (dates, publication types, scientific field)
- Combine **keywords** with **logical operators** (**AND**, **OR**, **NOT**) to refine the results
- Test different **formulations** & adjust the **queries** based on the **results obtained**

Automated Abstract of Scientific Articles

AI makes it possible to **quickly generate summaries** of **large articles** :

- **Identification** of **key points** and **main conclusions**
- **Creating** **comparative summaries** when several articles deal with the same subject
- **Saving time** for **researchers** wishing to focus on **critical analysis** rather than exhaustive reading

Critical Evaluation of AI Proposed Sources

Even though **AI** is very useful, it's **important** to verify the **quality** of the **sources**:

- Check the **reliability** and **credibility** of the **suggested articles**
- Analyze the **potential biases** of the **AI tool** (e.g., some platforms favor certain publishers or disciplines)
- Select only the **relevant** & **rigorous sources** for the **scientific project**

Conclusion

AI is transforming **information retrieval** by offering **powerful tools** for information gathering and analysis.

However, the researcher's role remains **crucial** in

- formulating the **right queries**,
- interpreting **results**, and
- selecting **reliable sources**.

By combining **AI** with a **critical approach**, it is possible to significantly improve the **quality** & **efficiency** of **scientific research**.

How use Elicit?

What is Elicit?

Elicit is an **AI Tool** that **enables** to :

- Retrieve Relevant Scientific Articles
- Abstract Publications
- Extract Key information :
 - * **Methodology**,
 - * **Results**,
 - * **Sample**,
 - * **etc.**
- Automatically compare many **studies**

Unlike Google Scholar, **Elicit** does more than just display links :

☞ it **analyses** **Contents** of **Articles** and Extracts **Important Information**

Steps of Utilisation

Step 1 : Formulate a Net Question

Elicit method works **best** with a **specific question**, and not just with **keywords**

Bad Example:



Artificial Intelligence education

Good Example:



What is the impact of artificial intelligence on student academic performance in higher education?

Step 2 : Launching the Research

While input the question :

 **Elicit** displays a List of **Relevant Scientific Articles**, generating a table containing :

- Title
- Year
- Type of study
- Methodology
- Main Results
- Automated Abstract

Step 3 : Analyzing & Comparing

We can:

- Add Columns (such as: sample size)
- Sort studies by date
- Comparing the results
- Export data

Concrete Example

Imagine you are preparing an Article :

The impact of deep learning on epidemiological surveillance

In **Elicit**, we put the following question :

How does deep learning improve infectious disease surveillance systems?

Article	Method	Data	Main Result
Study A (2023)	CNN	Twitter data	Early detection 2 weeks before official reports
Study B (2022)	LSTM	Hospital data	18% improvement in accuracy
Study C (2021)	Hybrid DL	Multi-source data	Reduction of the alert period

In some minutes, we obtain:



A Comparative view

Methodological trends

Calculated performance

The limitations mentioned

Without reading all 30 articles manually

Why it is Strong for Researchers ?



✓ Huge time saving

✓ Can Structure a Literature Review

✓ Enabling quick identification of gaps

✓ Ideal for preparing an introduction or a state-of-the-art review

Important Limitations

- ❑ ✓ **Elicit** does not replace reading the articles in their entirety
- ✓ It can summarize in simplified terms
- ✓ **It must always check the original sources**

PS:

Always Check the Original Sources

Practical Exercise
**Using Elicit for *AI*-assisted
scientific research**

Educational objectives

At the end of this exercise, the student will be able to :

- Formulate a precise research question
- Use Elicit to identify relevant scientific articles
- Compare multiple studies
- Summarize and critically analyze results
- Evaluate the reliability of sources suggested by AI

Part 1: Formulation of the question

Instructions

Choose a theme related to artificial intelligence

Examples of themes



AI and academic performance

Deep learning and public health

AI and climate change

Generative AI and education

Formulate a clear scientific question

PS: Formulate a clear scientific question

Formulate a clear scientific question

✗ **Bad example:**

▣ Artificial Intelligence in health

✓ **Good example**

▣ How does deep learning improve early detection of infectious diseases?

✍ **Task**

- ▣ • Write the question
- Justify why it is clear and precise (5–6 lines)

Part 2: Search with Elicit

Instructions

1. Log in to Elicit (www.Elicit.com)
2. Enter your question
3. Select the 5 most relevant articles

Complete the following table

Article	Year	Used Method	Data Type	Main Result	Limitations
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Part 3: Comparative Analysis

Answer the following questions :

- 1. Which AI method is most frequently used?**
- 2. Are the results consistent across studies?**
- 3. Are the sample sizes sufficient?**
- 4. Are there any contradictions?**
- 5. Which study seems the most robust? Why?**

(1-2 pages maximum)

Part 4: Critical evaluation of AI

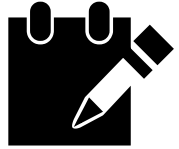
Answer :

- 1. Are the summaries generated by Elicit complete?**
- 2. Did you find any missing information after checking?**
- 3. Did Elicit suggest any irrelevant articles?**
- 4. Can the tool be trusted blindly? Justify your answer.**

Part 5: Mini-Summary

Write a 300–400 word summary:

- **Summarize the main trends**
- **Indicate the identified limitations**
- **Suggest a direction for future research**



Evaluation methods (example)

Criteria	Points
Quality of the question	/4
Relevance of the selected articles	/4
Comparative analysis	/6
Critical thionine	/4
Editorial quality	/2
Total	/20

Advanced Exercise

**Methodological Evaluation of
an AI Tool for Systematic
Review (Elicit)**

General objective

To scientifically evaluate the performance of Elicit in the context of a structured literature review and analyze its methodological limitations

Context

AI tools are now assisting literature searches.

But can they be considered reliable for rigorous scientific review?

Your mission is to conduct a comparative methodological mini-review.

Part 1: Protocol definition (research level)

Instructions

Choose a specific research question

Examples of themes

The question must be formulated according to a scientific structure :



- PICO (Population – Intervention – Comparison – Outcome)
- or
- SPIDER (qualitative)

Example

- In low-resource healthcare settings, does deep learning improve early detection of infectious diseases compared to traditional statistical models?

Required Tasks

- Justify the choice of wording
- Identify the keywords
- Define inclusion and exclusion criteria

Part 2: Comparative Search (Elicit vs. Traditional Method)

Step A: Search with Elicit

- Enter the complete query
- Export the first 15 articles returned
- Identify:
 - o Source database
 - o Year
 - o Journal impact factor
 - o Study type (experimental, review, meta-analysis, etc.)

Part 2: Comparative Search (Elicit vs. Traditional Method) (2)

• Step B: Manual Search

Perform the same search on:

- Google Scholar, or
- Scopus, or
- Web of Science, With:
 - Clear Boolean strategy
 - Identical time filtering

Part 3: Quantitative Analysis

Compare:

Criteria	Elicit	Classic search
Number of relevant articles		
Duplicates		
Off-topic articles		
Time required (minutes)		
Access to full text		

Statistical analysis required

- **Calculate the relevance rate**
- **Calculate the false positive rate**
- **Analyze thematic coverage**

Part 4: In-depth qualitative analysis

Answer the following questions:

- 1. Does Elicit bias the results towards certain disciplines?**
- 2. Is there a geographical bias?**
- 3. Are very recent articles correctly identified?**
- 4. Does the tool favor certain methodological approaches?**
- 5. Does the algorithm appear transparent?**

Part 5: Critical Audit of AI

Analyze:

- **Risk of algorithmic bias**
- **Risk of omitting major articles**
- **Reproducibility of results**
- **Problem of dependence on indexed databases**

Part 6: Scientific report (8–12 pages)

Expected structure:

- 1. Introduction**
- 2. Methodology**
- 3. Quantitative results**
- 4. Qualitative results**
- 5. Critical discussion**
- 6. Limitations**
- 7. Conclusion**

Mini-Projet : Revue de littérature assistée par l'Intelligence Artificielle

**Conception et évaluation d'une revue de littérature
assistée par IA : performance, biais et limites**

Context

AI tools like Elicit, Scopus AI, Web of Science Research Assistant, and ChatGPT are transforming literature searching

But:

- Are they reliable?
- Do they introduce biases?
- Can they replace a classic systematic methodology?

This mini-project aims to scientifically analyze these questions.

Educational Objectives

By the end of the project, the student should be able to:

- Formulate a structured research question
- Develop a scientific research strategy
- Use an AI tool for article retrieval
- Compare AI versus traditional methods
- Identify biases and limitations
- Produce a structured mini-scientific report

Assignment

Step 1: Defining the Research Question

Choose a topic related to:

- AI and public health
- AI and climate change
- AI and education
- AI and sustainability

Formulate the question using:

- PICO (quantitative), or
- SPIDER (qualitative)

Example:

Does deep learning improve early detection of infectious diseases compared to traditional machine learning methods?

Step 2: AI-Assisted Search

Use Elicit (or a similar tool).

1. Enter the complete search query
2. Select the first 15–20 articles
3. Export the results
4. Identify:
 - o Year
 - o Study type
 - o Journal
 - o DOI
 - o AI abstract

Step 3 : Classic Search

Reproduce the research with :

- . Google Scholar, ou
- . Scopus, ou
- . Web of Science

Example:

("deep learning" OR "neural networks")

AND ("infectious disease detection")

AND (early diagnosis)

Step 4 : Quantitative comparison

Construct a comparative table :

Criteria	AI	Classic search
Total number of items		
Relevant articles		
False positives		
Major articles missed		
Time needed		

Example:

Calculate:

- Relevance rate
- Error rate
- Time savings

Step 5: Critical Qualitative Analysis

:

Answer the following questions:

1. Does the AI favor certain types of journals?
2. Are recent articles adequately covered?
3. Is there a geographical bias?
4. Does the tool explain its sources?
5. Are the automatically generated abstracts accurate?

Mini-Project

AI-Assisted Literature Review in Public Health

**Conception et évaluation d'une revue de littérature
assistée par IA : performance, biais et limites**

Proposed Title

**Evaluation of AI Tools for Public Health
Literature Review: Application to Early
Screening of Infectious Diseases**

Scientific Context

Public health generates a massive volume of publications (epidemiology, modeling, digital surveillance, deep learning, IoMT, etc.)

AI tools such as:

- Elicit
- Scopus AI
- Web of Science Research Assistant

promise:

- time savings
- automatic summarization
- rapid identification of key articles

However, their methodological reliability remains poorly evaluated.

Problem statement

Can AI tools produce a reliable and reproducible literature review in public health?

Objectives

:

Students will be required to:

- Design a rigorous scientific research strategy
- Use an AI tool to collect articles
- Compare the results with traditional research
- Identify biases, omissions, and limitations
- Analyze the methodological impact on scientific quality

Research Question (Model)

PICO Formulation:

- Population: Populations exposed to emerging infectious diseases
- Intervention: Deep learning models
- Comparison: Traditional statistical methods
- Outcome: Improved early detection

Question: Does deep learning improve early detection of emerging infectious diseases compared to traditional epidemiological models?

Project Methodology

Step 1: AI-Assisted Search

Using Elicit:

1. Enter the full question
2. Extract the first 20 articles
3. Identify:
 - Year
 - Type of study
 - Country
 - Impact factor (if available)
 - DOI
 - AI-generated abstract

Step 2: Traditional Search

Perform the same search on:

- PubMed, or
- Scopus, or
- Web of Science

Example query:

("deep learning" OR "neural networks")
AND ("infectious disease detection")
AND ("early diagnosis" OR surveillance)

Filter:

- 2015–2026
- Scientific articles
- English

Comparative Analysis

Quantitative Analysis

Create a table:

Critère	IA	Recherche classique
Nombre total d'articles		
Articles pertinents		
Articles hors sujet		
Articles majeurs manqués		
Temps nécessaire (min)		

Calculate:

- Relevance rate
- False positive rate
- Time saved (%)

Bibliometric analysis

:

Compare:

- **Distribution by year**
- **Geographic distribution**
- **Types of journals (medical vs. computer science)**
- **Methodological types (experimental, review, meta-analysis)**

Critical Analysis

Students will discuss:

1. Disciplinary Bias

Does AI favor computer science articles over medical journals?

2. Time Bias

Are very recent articles included?

3. Geographic Bias

Is there overrepresentation of countries with high scientific output?

4. Transparency

Does the tool explain its sources?

5. Scientific Risk

Can AI alone be used for a systematic review?

Practical exercise

**Using Web of Science AI for a public
health literature review**

Learning Objectives

:

By the end of this exercise, students should be able to:

- Formulate a structured scientific question
- Use Web of Science AI to explore the literature
- Interpret an AI-generated synthesis
- Verify the quality of the results
- Identify methodological biases and limitations

Scientific Context

Emerging infectious diseases represent a major public health challenge. Artificial Intelligence models (Deep Learning, Machine Learning) are increasingly used for:

- epidemiological surveillance
- early detection
- outbreak prediction

P.S.: Web of Science AI offers intelligent exploration of indexed publications.

Part 1: Question Formulation

Formulate a question using the PICO model:

Example :

- Population: Populations exposed to emerging infectious diseases
- Intervention: Deep Learning
- Comparison: Classical statistical models
- Outcome: Improved early detection

Question:

Has deep learning significantly improved early detection of emerging infectious diseases over the past decade?

Tasks :

1. Identify the main keywords
2. Identify synonyms
3. Define the time period (e.g., 2015–2026)

Part 2: Research with Web of Science AI

Step 1 :

Enter the complete research question into Web of Science AI.

Step 2 :

Analyze:

- Generated summary
- Main thematic areas
- Key articles identified
- Time trends

Part 2: Research with Web of Science AI (2)

Step 3 :

Export the 15–20 main articles.

For each article, record:

- Year
- Journal
- Authors' country
- Number of citations
- Study type

Part 3: Quantitative Analysis

Construct a table:

Article	Year	Country	Citations	Type of Study	Relevant (Yes/No)
---------	------	---------	-----------	---------------	-------------------

Calculate:

- Relevance Rate (%)
- Average Citations
- Distribution by Year
- Geographic Distribution

Part 4: Verification via Classical Search

Perform an advanced query in Web of Science (classic mode):

Example:

```
TS=("deep learning" OR "neural networks")  
AND TS=("infectious disease" OR "epidemic prediction")  
AND TS=("early detection")  
Timespan: 2015–2026
```

Compare:

- Total number of articles
- Overlap with Web of Science AI
- Major articles not identified by the AI

Part 5: Critical Analysis

Students will discuss:

- 1. Is the AI-generated synthesis faithful to the articles?**
- 2. Does the AI favor highly cited articles?**
- 3. Is there a disciplinary bias (computer science vs. public health)?**
- 4. Are recent articles sufficiently represented?**
- 5. Can the tool replace a full systematic review?**

Part 6: Mini-Report (6–8 pages)

Expected Structure:

1. Introduction
2. Methodology
3. Results
4. Critical Discussion
5. Conclusion

Exercice comparatif

**Elicit vs Scopus AI pour une revue de
littérature en santé publique**

Objective

To compare the performance, scientific coverage, and potential biases of two AI tools:

- **Elicit**
- **Scopus AI**

within the context of research on:

Deep Learning for early detection of emerging infectious diseases.

Phase 1: Scientific Preparation

Research Question (PICO) :

- **Population:** Populations exposed to emerging infectious diseases
- **Intervention:** Deep Learning
- **Comparison:** Traditional statistical models
- **Outcome:** Early detection

Formulate a clear question and document:

- Main keywords
- Synonyms
- Time period (e.g., 2015–2026)

Phase 2: Search with Elicit

Procedure

1. Enter the complete question:

How does deep learning improve early detection of infectious diseases?

2. Extract the first 20 articles.

3. Export the metadata.

4. Note:

- o Year
- o Type of study
- o Domain
- o DOI
- o AI Abstract

Phase 3: Search with Scopus AI

Procedure

1. Enter the complete question.

How does deep learning improve early detection of infectious diseases?

2. Extract the first 20 articles.

3. Export the metadata.

4. Note:

- o Year
- o Type of study
- o Domain
- o DOI
- o AI Abstract

Export:

- 20 main articles
- Full metadata
- Number of citations
- Review

Phase 4: Comparative Analysis

Create a table:

Criteria	Elicit	Scopus AI
Total number proposed		
Really relevant articles		
False positives		
Search time		
Highly cited articles		
Recent articles (≤ 2 years old)		

Calculate:

- Relevance rate
- False positive rate
- % of recent articles
- Average citations

Overlap Analysis

Calculate:

- Number of common articles
- Articles unique to each tool

Similarity Index (Jaccard):

$$J = \frac{|A \cap B|}{|A \cup B|}$$

Critical Qualitative Analysis

Answer the questions:

1. Scientific Coverage

Does Scopus AI cover more medical journals?

2. Disciplinary Bias

Does Elicit favor computer science publications more?

3. Citation Bias

Does Scopus AI prioritize highly cited articles?

4. Abstract Quality

Are the AI summaries faithful to the articles?

5. Transparency

Which tool best explains its sources?

Expected results (Realistic Hypothesis)

In general:

- ♦ Scopus AI → better academic rigor, better traceability
- ♦ Elicit → faster, more exploratory, more flexible
- ♦ Scopus → favors highly cited articles
- ♦ Elicit → sometimes broader but less structured

Scientific Discussion

Points to analyze:

- Can Elicit alone be used for a systematic review?
- Is Scopus AI sufficient without a classic Boolean query?
- Should the two be combined?
- What impact on methodological quality in public health?

Scientific Discussion

Points to analyze:

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Expected Deliverable

8–12 page report:

Points to analyse:

- 1. Introduction**
- 2. Methodology**
- 3. Quantitative Results**
- 4. Qualitative Analysis**
- 5. Discussion**
- 6. Conclusion**

Statement:

- Adam likes apples.
- Inès likes carrots.
- Karim likes oranges.
- Apples are fruits.
- Oranges are fruits.
- Carrots are vegetables.
- Those who like fruit are healthy.

Questions:

1. Formalize these facts and rules in PROLOG.
2. What is the query to find out who is healthy?
3. What is the query to find out who likes apples?
4. How can we find out which fruits the program knows?

**Computer Science Department
First year of Bachelor's degree**

Prolog
Basic Concepts
COURSE of PROLOG: INTELLIGENCE
ARTIFICIELLE (IA)

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% --- Facts : who like what ---

Adam likes Apples.

like(adam, apples).

Ines likes carrots.

like(ines, carrots).

Karim likes oranges.

like(karim, oranges).

% --- Facts : categories ---

Apples are fruits.

fruit(apples).

Oranges are fruits.

fruit(oranges).

Carrots are vegetables.

vegetable(carrots).

% --- Facts : categories ---

fruit(apples).

fruit(orange).

vegetable(carrots).

% --- Rule : being healthy ---

healthy(Person) :- like(Person, Food),
 fruit(Food).

A person is healthy if he likes a food AND that food is a fruit.

🔍 Explanation of the rule

healthy(Person):-
like(Person, Food),
fruit(Food).

This means:

A person is healthy if he likes a food AND that food is a fruit.

✓ 2 □ Query: Who is healthy?

?- healthy(X).

✓ Expected result:

X = Adam;

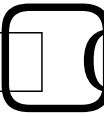
X = Karim.

Why?

Adam likes apples → fruit ✓

Karim likes oranges → fruit ✓

Inès likes carrots → vegetable ✗

✓ 3  Query: Who likes apples?

This allows us to ask:
?- food(X).

🔗 Pedagogical Summary

Prolog Query Question

Who is healthy? - healthy(X).

Who likes apples? - like(X, apples).

What are some fruits? - fruit(X).

?- likes(X, apples).

✓ Result:

X = Adam.

✓ 4 ☐ How to find out which fruits are known from the curriculum?

?- fruit(X).

✓ Result:

X = apples;

X = oranges.

🎓 Slightly more general version (optional)

We can also define:

food(X) :- fruit(X).

food(X) :- vegetable(X).